Chapter 10. The Lebesgue Integral

Lebesgue measure, measurable functions, convergence theorems, L1 and L2, Riesz–Fischer

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The Riemann integral handles continuous functions well but behaves badly under limits: pointwise limits of integrable functions need not be integrable, and the spaces $\mathcal{R}^1$, $\mathcal{R}^2$ are not complete (Example 7.2). Lebesgue's integral repairs both defects. We construct it on $\mathbb{R}$.

10.1 Measurable sets and measure

Proposition 10.1

Every open set $U \subset \mathbb{R}$ is a disjoint union of countably many open intervals, uniquely.

Proof. For $x \in U$ let I_x be the union of all open intervals containing x and contained in U; it is an open interval, and two such are equal or disjoint. Each contains a rational, so there are countably many. ∎

For an open set $U = \bigsqcup I_k$ put $m(U) = \sum |I_k|$, and for any $E \subset \mathbb{R}$ define the **outer measure**

$$m^*(E) = \inf\{m(U) : U \supset E \text{ open}\}.$$

Proposition 10.2

m^* is monotone, countably subadditive ($m^*(\bigcup E_n) \le \sum m^*(E_n)$), translation invariant, and $m^*(I) = |I|$ for intervals. Countable sets have outer measure 0.

Proof. Subadditivity: choose open $U_n \supset E_n$ with $m(U_n) < m^*(E_n) + \epsilon 2^{-n}$; then $\bigcup U_n$ is open and $m(\bigcup U_n) \le \sum m(U_n)$ (decompose the union into intervals and use compactness of closed subintervals). $m^*([a,b]) \ge b - a$ follows from the Heine–Borel theorem: an open cover has a finite subcover whose lengths sum to at least $b - a$. ∎

Definition (Lebesgue measurable set)

$E \subset \mathbb{R}$ is **measurable** if for every $\epsilon > 0$ there is an open $U \supset E$ with $m^*(U \setminus E) < \epsilon$. Then $m(E) = m^*(E)$. The class of measurable sets is denoted $\mathfrak{M}$.

Theorem 10.3

$\mathfrak{M}$ is a σ-algebra: $\emptyset \in \mathfrak{M}$; $E \in \mathfrak{M} \Rightarrow E^c \in \mathfrak{M}$; $E_n \in \mathfrak{M} \Rightarrow \bigcup E_n \in \mathfrak{M}$. It contains all open and closed sets and all sets of outer measure 0.

Sketch. Countable unions: choose $U_n \supset E_n$ with $m^*(U_n \setminus E_n) < \epsilon 2^{-n}$ and use subadditivity. Null sets: trivially measurable. Closed sets: a closed F is a countable union of compact sets, and for compact K and open $U \supset K$, $U \setminus K$ is open with $m(U \setminus K) = m(U) - m(K)$ (a computation with intervals). Complements: given E, choose open $U_n \supset E$ with $m^*(U_n \setminus E) < 1/n$; then $E^c = (\bigcup U_n^c) \cup Z$ with $Z = E^c \setminus \bigcup U_n^c \subset U_n \setminus E$ for all n, a null set, and $\bigcup U_n^c$ is a union of closed sets. ∎

Theorem 10.4 (Countable additivity)

If $E_n \in \mathfrak{M}$ are disjoint, then $m(\bigcup E_n) = \sum m(E_n)$.

Proof. $\le$ is subadditivity. For $\ge$, first for compact disjoint sets (they have positive distance, and an open cover of the union can be split), then for bounded measurable sets via inner approximation by compact sets (E measurable implies for each ϵ a closed $F \subset E$ with $m(E \setminus F) < \epsilon$), and finally in general. ∎

Consequently m is continuous along increasing sequences ($m(\bigcup E_n) = \lim m(E_n)$) and along decreasing sequences of finite measure. A property holds **almost everywhere** (a.e.) if it fails only on a null set. Not every set is measurable (Vitali's construction using a choice of representatives of $\mathbb{R}/\mathbb{Q}$), but every set that arises in practice is.

10.2 Measurable functions and the integral

Let $E \in \mathfrak{M}$ and $\mathbb{R}^* = [-\infty, \infty]$.

Proposition 10.5

For $f : E \to \mathbb{R}^*$ the following are equivalent, and define **measurability** of f: $\{f > a\} \in \mathfrak{M}$ for all $a \in \mathbb{R}$; $\{f \geq a\} \in \mathfrak{M}$ for all a; $\{f < a\} \in \mathfrak{M}$ for all a; $\{f \leq a\} \in \mathfrak{M}$ for all a.

Proof. $\{f \geq a\} = \bigcap_n \{f > a - \frac{1}{n}\}$, $\{f < a\} = \{f \geq a\}^c$, and so on. ∎

Continuous functions, monotone functions, sums, products, max, min, $\sup_n f_n$, $\limsup f_n$ and pointwise limits of measurable functions are measurable; changing a measurable function on a null set keeps it measurable. A **simple function** is a finite combination $s = \sum c_i \mathbf{1}_{A_i}$ of indicators of measurable sets.

Proposition 10.6

Every measurable $f : E \to [0, \infty]$ is the pointwise limit of an increasing sequence of nonnegative simple functions.

Proof. $s_n = \sum_{k=1}^{n2^n} \frac{k-1}{2^n} \mathbf{1}\{\frac{k-1}{2^n} \leq f < \frac{k}{2^n}\} + n\mathbf{1}\{f \geq n\}$. ∎

Definition (Lebesgue integral)

For simple $s = \sum c_i \mathbf{1}_{A_i} \geq 0$, $\int_E s = \sum c_i m(A_i \cap E)$. For measurable $f \geq 0$, $\int_E f = \sup\{\int_E s : 0 \leq s \leq f,\ s \text{ simple}\} \in [0, \infty]$. For real f with $\int |f| < \infty$ (written $f \in L^1(E)$), $\int f = \int f^+ - \int f^-$; for complex f, integrate real and imaginary parts.

Theorem 10.7

If $f \geq 0$ is measurable on $E = \bigsqcup A_n$ (disjoint measurable), then $\int_E f = \sum_n \int_{A_n} f$. If $f \geq 0$ and $\int_E f = 0$, then $f = 0$ a.e.

Proof. For simple f this is countable additivity of m; in general take suprema. For the second claim, $m\{f > 1/n\} \leq n \int f = 0$ (Chebyshev). ∎

10.3 Convergence theorems

Theorem 10.8 (Monotone convergence)

If $0 \leq f_1 \leq f_2 \leq \cdots$ are measurable and $f_n \to f$ pointwise, then $\int f_n \to \int f$.

Proof. Clearly $\lim \int f_n \leq \int f$. For simple $0 \leq s \leq f$ and $c < 1$, the sets $E_n = \{f_n \geq cs\}$ increase to E, so $\int f_n \geq \int_{E_n} f_n \geq c \int_{E_n} s \to c \int s$ (Theorem 10.7). Let $c \to 1$ and take the supremum over s. ∎

Consequences: the integral is additive on nonnegative functions (approximate by simple functions), and $\int \sum f_n = \sum \int f_n$ for $f_n \geq 0$.

Theorem 10.9 (Riemann versus Lebesgue)

A bounded $f : [a, b] \to \mathbb{R}$ is Riemann integrable iff the set of its discontinuities has measure zero, and then f is Lebesgue integrable with the same integral. Absolutely convergent improper Riemann integrals are Lebesgue integrals.

Sketch. Take partitions P_k refining each other with mesh $\to 0$; the step functions l_k, u_k built from m_i, M_i increase/decrease to limits $l \le f \le u$ with $l = u$ exactly at the continuity points of f (off the countable set of partition points). By monotone convergence $\int l = \lim L(f, P_k)$ and $\int u = \lim U(f, P_k)$, so f is Riemann integrable iff $\int (u - l) = 0$ iff $u = l$ a.e. ■

So the Dirichlet function, not Riemann integrable, has Lebesgue integral 0.

Theorem 10.10 (Fatou's lemma)

For measurable $f_n \ge 0$, $\int \liminf f_n \le \liminf \int f_n$.

Proof. Apply monotone convergence to $g_n = \inf_{k \ge n} f_k \uparrow \liminf f_n$, noting $\int g_n \le \int f_k$ for $k \ge n$. ■

Theorem 10.11 (Dominated convergence)

If $f_n \to f$ a.e., $|f_n| \le g$ and $g \in L^1(E)$, then $f \in L^1$ and $\int |f_n - f| \to 0$; in particular $\int f_n \to \int f$.

Proof. Apply Fatou to $2g - |f_n - f| \ge 0$: $\int 2g \le \liminf \int (2g - |f_n - f|) = \int 2g - \limsup \int |f_n - f|$. ■

Theorem 10.12 (Beppo Levi)

If $f_n \in L^1(E)$ and $\sum \int |f_n| < \infty$, then $\sum f_n$ converges absolutely a.e., its sum is in L^1, and $\int \sum f_n = \sum \int f_n$.

Proof. $g = \sum |f_n|$ has $\int g = \sum \int |f_n| < \infty$ by monotone convergence, so $g < \infty$ a.e.; apply dominated convergence to the partial sums, dominated by g. ■

Example 10.1

(a) Dominated convergence explains Example 6.1(b): $nx(1-x^2)^n$ has no integrable majorant.
(b) $\int_0^\infty \frac{x}{e^x - 1} dx = \sum_{n \ge 1} \int_0^\infty x e^{-nx} dx = \sum \frac{1}{n^2} = \frac{\pi^2}{6}$, by monotone convergence.
(c) Differentiation under the integral sign holds whenever $|\partial_x f(x, t)| \le g(t)$ with $g \in L^1$ (mean value theorem plus dominated convergence), a much more flexible version of Theorem 8.3.

Theorem 10.13 (Absolute continuity)

If $f \in L^1(E)$, then for every $\epsilon > 0$ there is $\delta > 0$ with $\int_A |f| < \epsilon$ whenever $A \subset E$ is measurable with $m(A) < \delta$.

Proof. $\int |f| \mathbf{1}\{|f| > n\} \to 0$ by dominated convergence; choose n with this $< \epsilon/2$ and $\delta = \epsilon/(2n)$. ■

10.4 The spaces L^1 and L^2; Fourier series again

Identify functions equal a.e. Then $\|f\|_1 = \int |f|$ is a norm on $L^1(E)$, and $\|f\|_2 = (\int |f|^2)^{1/2}$ on $L^2(E)$, with inner product $\int f\bar{g}$.

Theorem 10.14 (Completeness)

$L^1(E)$ and $L^2(E)$ are complete; moreover every Cauchy sequence has a subsequence converging a.e.

Proof. By Proposition 7.8 it suffices that absolutely convergent series converge. For L^1 this is Beppo Levi's theorem. For L^2, if $\sum \|f_n\|_2 < \infty$, then $g = \sum |f_n|$ satisfies $\|g\|_2 \le \sum \|f_n\|_2$ (Minkowski plus monotone convergence), so $\sum f_n$ converges a.e. and, dominated by g^2, in L^2. ■

For periodic functions the theory of Chapter 9 extends verbatim from $\mathcal{R}^1, \mathcal{R}^2$ to L^1, L^2 on $[-\pi, \pi]$: Riemann–Lebesgue, Fejér's theorem in the L^1 norm, and Parseval's identity. Completeness of L^2 adds the converse:

Theorem 10.15 (Riesz–Fischer)

For every $c \in \ell^2(\mathbb{Z})$ there is $f \in L^2[-\pi, \pi]$ with $\hat{f}(n) = c(n)$ for all n. Thus $f \mapsto \hat{f}$ is an isometric isomorphism of $L^2[-\pi, \pi]$ (normalized measure) onto $\ell^2(\mathbb{Z})$.

Proof. The partial sums $P_N = \sum_{|n| \le N} c(n) e^{inx}$ form a Cauchy sequence in L^2 (orthonormality), so they converge to some f; then $\hat{f}(n) = \lim_N \langle P_N, e_n \rangle = c(n)$ by continuity of the inner product. Injectivity is Parseval. ■

Theorem 10.16 (Uniqueness)

If $f \in L^1[-\pi, \pi]$ and $\hat{f}(n) = 0$ for all n, then $f = 0$ a.e.

Proof. Fejér means $\sigma_N(f) = 0$ converge to f in L^1 (approximate f by continuous functions and use $\|\sigma_N(g)\|_1 \le \|g\|_1$, since $K_N \ge 0$ has integral 1). ■

Theorem 10.17 (Density)

Continuous functions with compact support (and step functions) are dense in $L^1(E)$ and in $L^2[a, b]$.

Proof. Simple functions are dense (Proposition 10.6 and dominated convergence); a measurable set of finite measure is approximated in measure by an open set, hence by a finite union of intervals, whose indicator is approximated by continuous functions. ■

10.5 Function spaces and Fourier coefficients

The correspondence $f \mapsto \hat{f}$ links function spaces with sequence spaces:

analytic periodic functions	$\leftrightarrow$	$\{a : \limsup \lvert a(n)\rvert^{1/\lvert n\rvert} < 1\}$
C^∞ periodic functions	$\leftrightarrow$	rapidly decreasing sequences
continuous periodic functions	$\to$	(a proper subset of) $\ell^2(\mathbb{Z})$
$L^2[-\pi, \pi]$	$\leftrightarrow$	$\ell^2(\mathbb{Z})$
$L^1[-\pi, \pi]$	$\to$	(a proper subset of) $c_0(\mathbb{Z})$

The map $L^1 \to c_0$ is injective (Theorem 10.16) but not onto (Section 9.4). Moreover, the coefficients of an L^1 function need not be square-summable, as the following shows.

Example 10.2 (Periodization)

For $g \in L^1(\mathbb{R})$, $G(x) = \sum_{n\in\mathbb{Z}} g(x - 2n\pi)$ converges a.e. and defines $G \in L^1[-\pi, \pi]$ with $\int_{-\pi}^{\pi}|G| \le \int_{\mathbb{R}}|g|$ (monotone convergence), and dominated convergence gives

$$\hat{G}(n) = \frac{1}{2\pi}\int_{-\infty}^{\infty} g(x)e^{-inx}dx = \frac{1}{2\pi}\hat{g}(n).$$

Choosing $g \in L^1(\mathbb{R})$ whose Fourier transform is not square-summable on $\mathbb{Z}$ (e.g. $g(x) = |x|^{-1/2}\mathbf{1}\{|x| \le 1\}$, for which $\hat{g}(n) \sim c|n|^{-1/2}$) produces $G \in L^1$ with $\hat{G} \notin \ell^2$. The same identity, combined with Fourier inversion, yields the Poisson summation formula $\sum g(2n\pi) = \frac{1}{2\pi}\sum \hat{g}(n)$ for nice g.